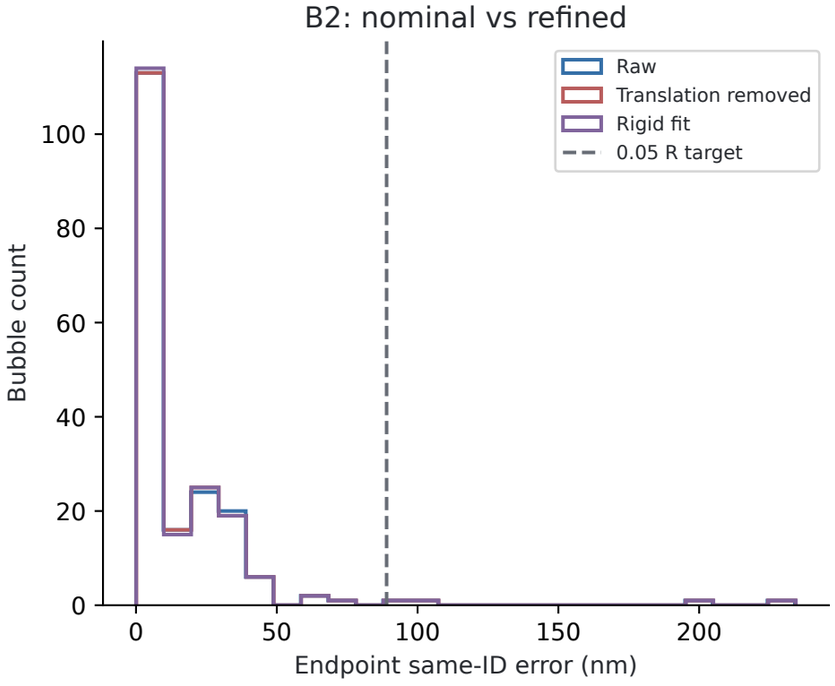
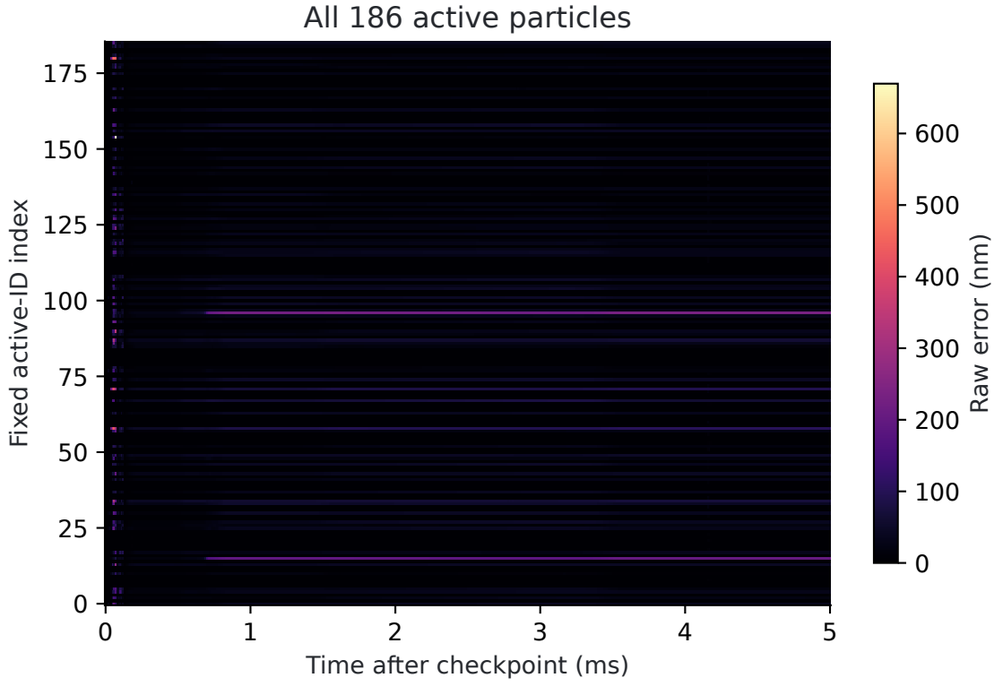
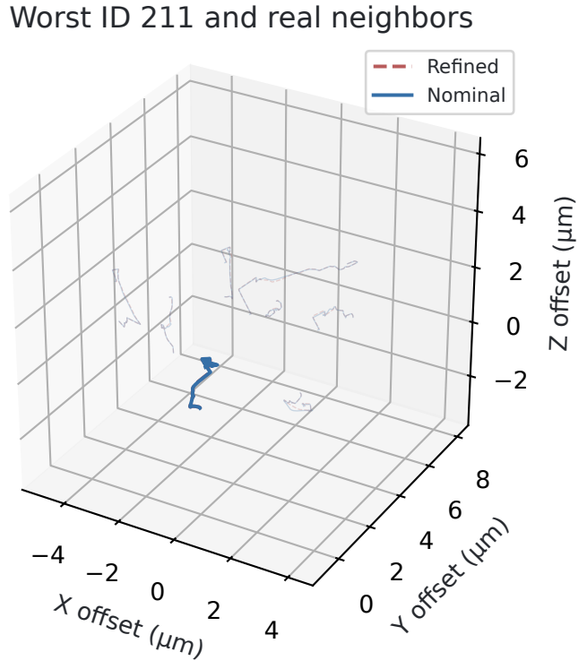
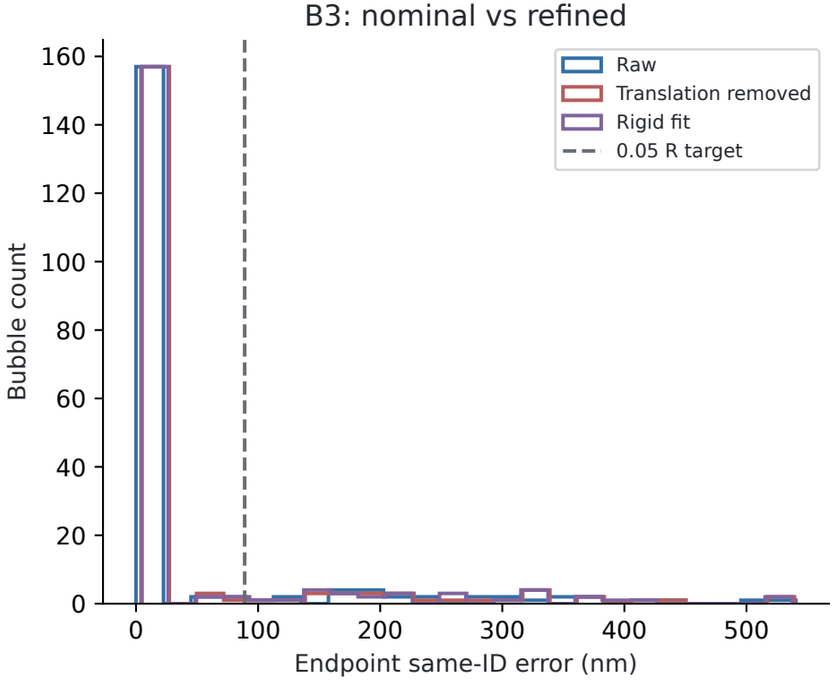
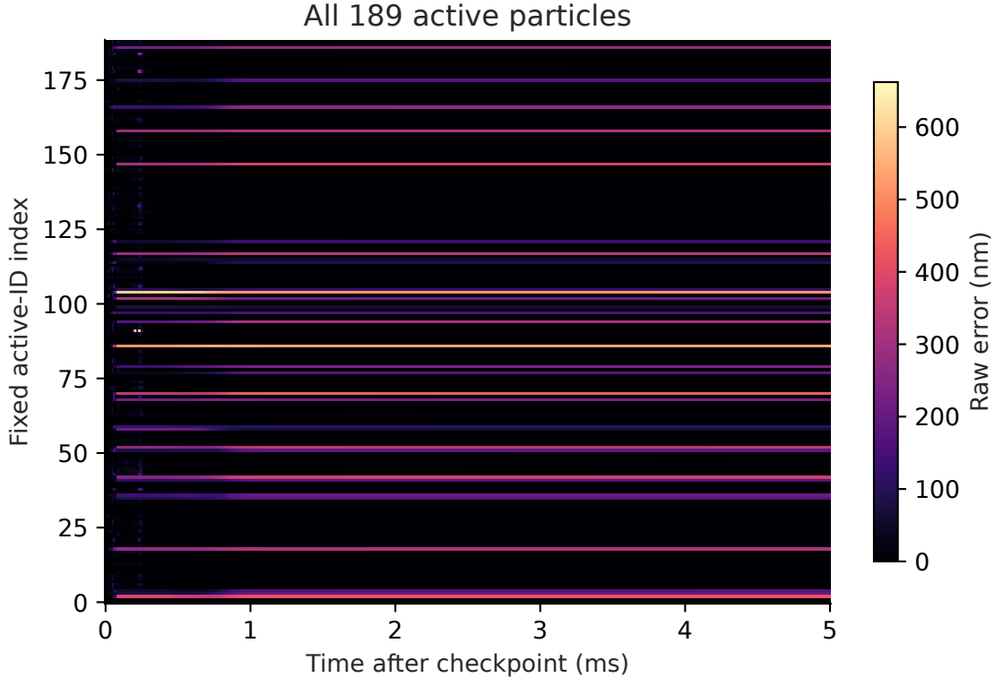
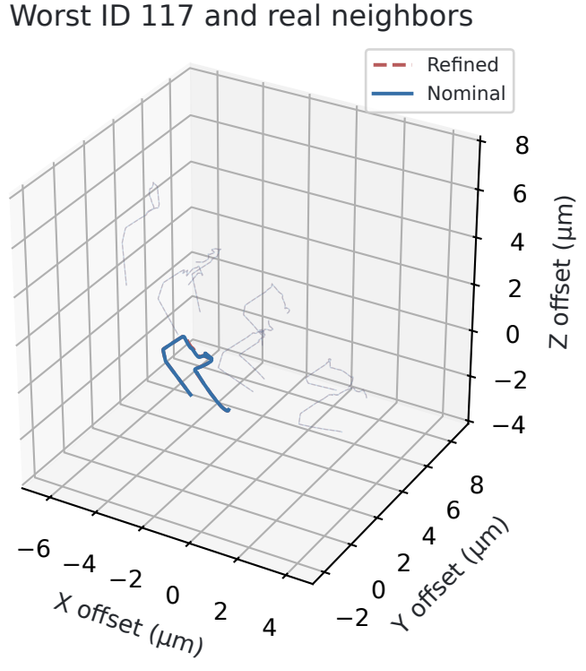

B2: nominal vs refined
Raw
Translation removed
Rigid fit
0.05 R target
Bubble count
Endpoint same-ID error (nm)


All 186 active particles
Fixed active-ID index
Time after checkpoint (ms)
Raw error (nm)


Worst ID 211 and real neighbors
Refined
Nominal
X offset (μm)
Y offset (μm)
Z offset (μm)


B3: nominal vs refined
Raw
Translation removed
Rigid fit
0.05 R target
Bubble count
Endpoint same-ID error (nm)


All 189 active particles
Fixed active-ID index
Time after checkpoint (ms)
Raw error (nm)


Worst ID 117 and real neighbors
Refined
Nominal
X offset (μm)
Y offset (μm)
Z offset (μm)